# Showkot Hossain

Ph.D. Student, Computer Science and Engineering, University of Notre Dame

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## Research Interests

Security and privacy, distributed systems, confidential computing

## Education

Aug 2024 – Present — **Ph.D. in Computer Science and Engineering** — *Notre Dame, IN, USA*
*University of Notre Dame*
Advisor: Dr. Taeho Jung · GPA: 3.91/4.0

Jan 2016 – Jan 2020 — **B.Sc. in Computer Science and Engineering** — *Dhaka, Bangladesh*
*University of Dhaka*
Thesis: Network Log Management System using Blockchain
Advisors: Dr. Mosarrat Jahan and Dr. Upama Kabir

## Publications

*Conference Papers*

[C6] **Decentralized AI Governance Must Decouple Policy Processing from Capability Enforcement**
Hasan Kassem, Christoforos Anagnostopoulos, Alejandro Aristizabal, Spyridon Bakas, Orion GB Banks, Omar Benjelloun, Tian Cai, Sergen Cansiz, Davide Chicco, Patrick Foley, **Showkot Hossain**, Taeho Jung, Peter Kairouz, Yanchen Liu, Marco Lorenzi, Peter Mattson, Ann K Novakowski, Michael O'Connor, Holger R Roth, Adrish Sannyasi, Charalampos Savvaidis, Micah Sheller, David Solooki, Dimitris Stripelis, Renato Umeton, Marc Vesin, Ray Wang, Wenbin Zhang, Mic Bowman, Alexandros Karargyris
*40th Annual Conference on Neural Information Processing Systems (NeurIPS 2026), Position Paper Track.*
Acceptance rate: 22.9%

[C5] **Cost-Effective Logging for Decentralized Massive Data Sharing**
Avery Hughes, **Showkot Hossain**, Yanchen Liu, Changhao Chenli
*9th IEEE International Conference on Blockchain (IEEE Blockchain 2026)*

[C4] **Web3IoT: A Decentralized Database Engine for Secure and Scalable IoT Data Management**
Omkar Rokdoba Doijad, Chandan Monabolu Narayana, **Showkot Hossain**, Jiannong Cao, Taeho Jung, Haijian Sun, WenZhan Song
*IEEE International Conference on E-health Networking, Application & Services (IEEE HealthCom 2026)*

[C3] **MtDB: A Decentralized Multi-Tenant Database for Secure Data Sharing**
**Showkot Hossain**, Wenyi Tang, Changhao Chenli, Haijian Sun, WenZhan Song, Seokki Lee, Mic Bowman, Taeho Jung
*8th IEEE International Conference on Blockchain (IEEE Blockchain 2025), pp. 79–86*

[C2] **SecLog: Atomic-Swap Based Secure Logging for Decentralized Data Sharing**
Avery Hughes, **Showkot Hossain**, Wenyi Tang, Taeho Jung, Changhao Chenli
*International Conference on Metaverse (METAVERSE 2025), Springer, pp. 115–124*

[C1] **Towards Blockchain-Driven Network Log Management System**
MH Rakib, **Showkot Hossain**, Mosarrat Jahan, Upama Kabir
*IEEE 8th International Conference on Smart City and Informatization (iSCI 2020)*

*Journal Articles*

[J2] **Consentful-by-Design: A Perspective on Safeguarding Data Ownership from Generative AI Leveraging Lessons from the Healthcare Domain**
Akis Linardos, Theresa Willem, Alena Buyx, **Showkot Hossain**, Taeho Jung, Dimitrios Makris, Spyridon Bakas
*AI & SOCIETY, Springer, pp. 1–13, 2026*

[J1] **A Blockchain-Enabled Scalable Network Log Management System**
MH Rakib, **Showkot Hossain**, Mosarrat Jahan, Upama Kabir
*Journal of Computer Science, 18(6), pp. 496–508, 2022*

## Research Experience

Aug 2024 – Present
**Graduate Research Assistant** *Notre Dame, IN, USA*
*Data Security and Privacy Lab (DSP-Lab), University of Notre Dame*
Advisor: Dr. Taeho Jung
- Research and develop Web3DB, a decentralized platform for secure data sharing and management, and Web3FS, its decentralized file system, as part of a collaborative team; published several conference papers.

Jan 2020 – Dec 2020
**Research Assistant** *Dhaka, Bangladesh*
*Cyber Security Lab, University of Dhaka*
Advisors: Dr. Mosarrat Jahan and Dr. Upama Kabir
- Built a prototype system for secure provenance tracking and real-time monitoring using PKI and blockchain; published one peer-reviewed journal article and one conference paper.

## Industry Experience

Nov 2023 – Jul 2024
**Software Engineer, Level 3** *Dhaka, Bangladesh*
*Itransition Software Bangladesh Limited*
- Collaborated with global clients to design and implement high-performance APIs for Product Madness, a leading mobile game studio with 5M+ monthly active users, consistently meeting release deadlines.
- Led a team of 5 developers building and maintaining features with Spring Boot (Java), React, and PostgreSQL in an Agile process (JIRA, Git, JUnit, Mockito).

Mar 2023 – Nov 2023
**Software Engineer, Level 2** *Dhaka, Bangladesh*
*Kona Software Lab Limited*
- Implemented the authentication and authorization system for microservices serving 13M+ KonaCard users.
- Redesigned the database architecture and wrote a migration script, reducing query latency by 25%.

Oct 2020 – Mar 2023
**Software Engineer, Level 1** *Dhaka, Bangladesh*
*Dohatec New Media*
- Researched and developed a microservices-based backend for ePMIS, the Government of Bangladesh's Electronic Project Management Information System, built for high performance and scalability.
- Designed and developed high-performance RESTful APIs with Spring (Java), AngularJS, and SQL Server in cross-functional Agile teams.

Jun 2020 – Aug 2020
**Blockchain Developer (Intern)** *Dhaka, Bangladesh*
*Technohaven Company Limited*
- Studied blockchain architecture, consensus algorithms, digital identity, and blockchain applications.
- Developed a prototype supply-chain traceability system using Hyperledger Fabric [code].

## Honors, Awards & Grants

2025 **Travel Grant**, University of Notre Dame. $1,532 to present at IEEE Blockchain 2025 (Zhengzhou, China).

2024 **Travel Grant**, University of Notre Dame. $1,000 to attend ACM CCS 2024 (Salt Lake City, UT, USA).

2024 – 2029 **Research Assistantship**, University of Notre Dame. Full tuition, health insurance, and stipend.

2020 – 2021 **ICT Innovation Fund**, ICT Division, Government of Bangladesh. $5,000 grant for innovative project development.

2020 **Award of Merit**, International Blockchain Olympiad (team of 4).

2020 **9th Place**, Blockchain Olympiad Bangladesh (team of 4).

2011 – 2020 **Government Education Board Scholarships**, Bangladesh. Awarded three times for national examination results: top 1% in the Higher Secondary Certificate (2015), top 2% in the Secondary School Certificate (2013), and top 1% in the Junior School Certificate (2010).

## Teaching

Fall 2025 **Teaching Assistant**, CSE 20110: Discrete Mathematics, University of Notre Dame

## Mentoring

Undergraduate researchers, co-supervised with Dr. Taeho Jung.
*University of Notre Dame · †Indiana Institute of Technology

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|---|---|
| Spring 2026 | Will Armsworthy*, Paola Lopez*, Aaron Wang* |
| Fall 2025 | Will Armsworthy*, Maria C. Leal*, Avery Hughes† |
| Spring 2025 | Ryan Nahm*, Aaron Wang*, Avery Hughes† |
| Fall 2024 | Gustavo Aniceto*, Jonathan D'Alonzo* |

## Professional Service

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| 2026 | Reviewer, ACM Distributed Ledger Technologies (DLT) |
| 2026 | Reviewer, IEEE Transactions on Dependable and Secure Computing (TDSC) |
| 2026 | Reviewer, IEEE Network |
| 2025 – 2026 | Reviewer, IEEE International Conference on Blockchain and Cryptocurrency (ICBC) |
| 2025 | Reviewer, IEEE International Conference on Blockchain (Blockchain) |

## Leadership & Outreach

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| 2025 – 2026 | **Sports & Cultural Secretary**, Bangladeshi Students Association of Notre Dame (BDSA-ND) |
| 2017 – 2024 | **Co-founder**, Jiboner Alo. Education for orphaned and underprivileged children. |
| 2016 – 2021 | **Tutor**, private tutoring for high school and undergraduate students. |
| 2016 – 2020 | **Member**, Badhan (Dr. Muhammad Shahidullah Hall, University of Dhaka). Organized blood donation drives and educational workshops. |
| 2019 | **Student Volunteer**, International Conference on Innovation and Engineering (ICIET) |
| 2019 | **Mentor**, SCB-PA Inter School and College Programming Contest (ISCPC) |

## Technical Skills

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| Languages | Python, Java, C/C++, JavaScript, Solidity |
| Systems & Web | Spring Boot, React, Node.js, FastAPI, Parse, JPA, Hibernate, RabbitMQ, Kafka, ELK Stack, Gradle, JUnit, Mockito |
| Databases | PostgreSQL, Microsoft SQL Server, MongoDB, Redis |
| ML | Pandas, NumPy, PyTorch, TensorFlow |
| Blockchain | Smart contracts (Solidity), Hyperledger Fabric |
| Tools | Trusted Execution Environments (TEE), Docker, Jenkins, Nexus, Postman, Swagger, JMeter, Jira, Git |

## Selected Projects

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| PostStream | A micro-blogging application inspired by Twitter, built with React and Parse Server, with real-time content discovery, social interactions, and dual-platform deployment [code]. |
| Emotion | A BERT-based multi-class classifier for 13 tweet emotions, improving accuracy on an imbalanced dataset through synonym augmentation, ChatGPT-generated synthetic data, and class filtering [code]. |
| Spam Detector | A DistilBERT-based spam detector on the Enron dataset reaching 99.36% accuracy with balanced precision and recall, outperforming traditional ML and deep-learning baselines [code]. |

## References

**Dr. Taeho Jung**
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University of Notre Dame, IN, USA
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*Ph.D. advisor*

**Dr. WenZhan Song**
Professor
School of Electrical and Computer Engineering
University of Georgia, Athens, GA, USA
wsong@uga.edu
*Research collaborator*